

SECTIONS - About You, Additional Contact Details, Academic Program Details

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Research Involvement Details

First name: Neil

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Do you have additional WSU research mentors? N/A (don't check anything)

Number of semesters in research: 2

Number of summers in research: 0

About Your SURCA Presentation

Is this a join presentation: Yes

Names of all co-authors [exclude yourself, keep our client's name at the end]:

Alexia Jackson

Angelica Vargas

David Villagomez

Tucker Brisbois

Martha Cárdenas

Presentation Method: I will come to Pullman to present my poster

Do you have any special needs? Fill this in on your own

About the Project You are Presenting

Project category: Computer Science, Mathematics, Statistics, and Information Sciences

Project title: Operación Salud Colima Tamizaje

Project Abstract:

Operación Salud Colima Tamizaje, also known as the Colima Health Screening Operation, is an innovative web-based computer application designed to transform the health screening process

for children in Colima, Mexico. This application was developed as part of a senior capstone design project by a team of four computer science students at Washington State University Tri-Cities. The project addresses significant challenges in managing health records, unclear scheduling of medical visits, misplaced health paperwork, and limited communication between schools, medical professionals, and parents.

The application addresses these issues by offering digital consent forms that parents can sign and submit electronically, eliminating the need for physical paperwork while ensuring an organized and secure storage of forms. Parents also have the option to download copies of these forms. Furthermore, the application provides parents with instant access to a comprehensive range of their child's health data, including glucose levels, cholesterol, and other important health metrics. It includes a schedule of health service visits to schools, along with downloadable lab results and health recommendations, centralizing all critical information in one location. This functionality enables parents to effectively monitor their child's health progress.

To further enhance the system's usability, administrators and medical professionals can update and modify student information, ensuring that parents have immediate access to the most up-to-date and accurate health data. The application streamlines data collection and reporting, ensuring effective communication of health insights across all stakeholders.

Developed in collaboration with the Ministry of Health and Social Welfare of Colima, the system adheres to regional health regulations and user needs. Modern software engineering principles guided the development of this system, utilizing PHP and Laravel for back-end development, MySQL for secure data storage, and Bootstrap for a user-friendly front-end interface. The application integrates with Excel-based health records to automate data imports, reducing errors and improving efficiency.

By automating repetitive tasks and optimizing data flow between stakeholders, this web application significantly reduces administrative burdens for schools and healthcare providers in Colima, Mexico. Ultimately, Operación Salud Colima Tamizaje serves as a robust tool that enhances public health monitoring and contributes to early intervention in children's health issues.